LOCAL COHOMOLOGY INTERACTS WITH TENSOR PRODUCTS THROUGH THE REPRESENTING OBJECT

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- The description of local cohomology as tensoring with a fixed object. Invoked at the start of the counted proof (source lines 1640-1643). Provides the representing object whose tensor properties give the assertion.

Source reference. Original lemma lemma-local-cohomology-adjoint: sources/stacks-dualizing/pinned-original-source.tex, source lines 1466–1466. Navigation pointer only: the counted text refers to the Stacks item lemma-local-cohomology-adjoint, which is an existing labeled result of this same chapter and is not part of this card. The pointer carries the target's own label as its number and its definition line as its span; its content is not reproduced here.

Lemma 0.1. *Let A be a ring and let $I \subset A$ be a finitely generated ideal. For $K, L \in D(A)$ we have*

$$R\Gamma_Z(K \otimes_A^{\mathbf{L}} L) = K \otimes_A^{\mathbf{L}} R\Gamma_Z(L) = R\Gamma_Z(K) \otimes_A^{\mathbf{L}} L = R\Gamma_Z(K) \otimes_A^{\mathbf{L}} R\Gamma_Z(L)$$

If K or L is in $D_{I^\infty\text{-}torsion}(A)$ then so is $K \otimes_A^{\mathbf{L}} L$.

Proof. By Lemma lemma-local-cohomology-adjoint we know that $R\Gamma_Z$ is given by $C \otimes^{\mathbf{L}} -$ for some $C \in D(A)$. Hence, for $K, L \in D(A)$ general we have

$$R\Gamma_Z(K \otimes_A^{\mathbf{L}} L) = K \otimes^{\mathbf{L}} L \otimes_A^{\mathbf{L}} C = K \otimes_A^{\mathbf{L}} R\Gamma_Z(L)$$

The other equalities follow formally from this one. This also implies the last statement of the lemma. □

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